Source Attribution of Watermarked Text

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Motivation & Problem

Multiple LLMs embed cryptographic watermarks for provenance. When an observer receives text, which model produced it? Two attribution paradigms exist:

Watermark (Active)
- Per-model KGW key
- z-test each key
- Strong on clean text

Fingerprint (Passive)
- TF-IDF n-gram features
- 4-way classifier
- No key needed

Research Gaps

1. Does watermarking compress inter-model distributional differences?
2. Head-to-head watermark vs. fingerprint attribution under attack?
3. Can sentence-level watermarks (SimMark) complement token-level (KGW)?

Method — KGW Watermarking

Kirchenbauer et al., ICML 2023. Token-level green/red partition with logit bias. Detection via z-test on green token count.

```
Z = (|s|_G − □T) / √(□(1−□)T)
```

Parameters:
`□ = 0.25 · □ = 2.0 · z-threshold = 3.0`

Models:
LLaMA-3.2-1B, LLaMA-3.2-3B, Gemma-2B, Phi-2

Attacks:
Synonym sub. 10–70%, Ins/Del 10–30%, Reorder, Neural (Qwen-3)

F1 Distributional Compression

Watermarking shrinks inter-model differences to 1/3 of clean baseline (Gemma-2B vs Phi-2).

Setting	Wt.Diff	Ent.Diff
Indep. WM keys	0.24	0.76
No WM (clean)	**0.76**	**2.30**
Shared green list	**1.56**	**4.48**

Green-list bias dominates each model's natural signature — a direct headwind for passive fingerprint attribution.

F2 Robustness Crossover

Two paradigms fail in non-overlapping regimes. Watermark: cliff-like collapse. Fingerprint: graceful degradation. Lines cross at 50–70% synonym substitution.

Synonym %	Watermark	Fingerprint	Winner
0% (clean)	**96.7%**	76.7%	Watermark
10–30%	**93–97%**	70–77%	Watermark
50%	73.3%	66.7%	Watermark
70%	**13.3%**	**63.3%**	**Fingerprint**

96.7%
WM clean accuracy

13.3%
WM @ 70% synonym

63.3%
FP @ 70% synonym

Takeaway: production attribution should combine both — failure modes are non-overlapping.

F3 Hybrid KGW + SimMark

SimMark (EMNLP 2025): sentence-level cosine similarity watermark. Combined with KGW via rejection sampling. Detection: OR logic — either positive watermarked.

SimMark vs KGW — Moderate Paraphrase (5 rounds)

Metric	SimMark	KGW
Survival Rate	**80%**	**20%**
z-score Decay	34%	**75%**

Hybrid Robustness (OR-logic detection)

Scenario	KGW	SimMark	Hybrid OR
Short, no atk	66.7%	33.3%	66.7%
Poetry, no atk	66.7%	66.7%	**100%** □
Long, mod. atk	0%	**100%**	**100%**
Any, aggressive	0%	0%	0%

KGW Rescues
- Short text (≤3 sent.): SimMark lacks stats
- Haiku: KGW z=5.89, SimMark z=−0.58

SimMark Rescues
- Moderate rewrite: semantics preserved
- Long: SimMark z=3.87, KGW z=1.66

F4 Multi-LLM Chain Paraphrase

17 experiments: Llama-3.2-3B→Qwen-3 paraphraser. KGW watermarks do not survive neural paraphrase.

17
chain experiments

6%
survival (1/17)

5.07
avg z-decay

Mode	Orig z	After z	Survived
Standard	5.73	−1.16	No
Creative	5.73	0.77	No
Structure	5.73	0.05	No
Summary-exp.	5.73	−0.20	No

Neural paraphrase is far more destructive than symbolic synonym substitution.

Conclusions

1. Watermarking compresses distributions — makes different models look more alike.
2. Watermark & fingerprint: non-overlapping failure modes (crossover at 50–70%).
3. KGW + SimMark are complementary — Hybrid OR achieves 100% on poetry.
4. Neural paraphrase defeats KGW alone — only 1/17 survived.
5. Aggressive attack remains an open challenge for all current methods.

Future Work

- → Hybrid attribution classifier: KGW z-scores + TF-IDF features in one model
- → Neural paraphrase attacks: extend with GPT/Claude rewrite
- → Efficient hybrid generation: sentence-level beam search (SentBS)
- → Adaptive SimMark intervals auto-tuned per text type

References

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Website: creegon.github.io/Design_Project
Code: github.com/creegon/Design_Project